Week 5 Tutorial Solutions

Problem 19

Find the slope of the tangent line to the curve $y^2 + xy - y^3 = 2$ at the point $(3, 2)$.

Solution

Check the point. Substituting $(x, y) = (3, 2)$ gives $2^2 + (3)(2) - 2^3 = 4 + 6 - 8 = 2$, so the point lies on the curve.

Step 1: Differentiate implicitly with respect to x. Treating y as a function of x, we apply the chain rule to the y terms and the product rule to xy:

$$2y\,\frac{dy}{dx} + \left(y + x\,\frac{dy}{dx}\right) - 3y^2\,\frac{dy}{dx} = 0.$$

Step 2: Collect the $\frac{dy}{dx}$ terms.

$$\frac{dy}{dx}\left(2y + x - 3y^2\right) = -y \quad \Longrightarrow \quad \frac{dy}{dx} = \frac{-y}{2y + x - 3y^2}$$

Step 3: Substitute $(x, y) = (3, 2)$.

$$\left.\frac{dy}{dx}\right|_{(3,2)} = \frac{-2}{2(2) + 3 - 3(2)^2} = \frac{-2}{4 + 3 - 12} = \frac{-2}{-5} = \frac{2}{5}$$

Answer. The slope of the tangent line at $(3, 2)$ is

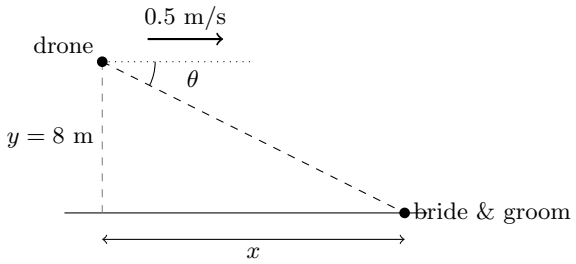
$$\boxed{\frac{dy}{dx} = \frac{2}{5}}.$$

Problem 20

A wedding photographer is using an aerial drone to film a wedding. She keeps the drone at a steady height of 8 m above the ground, moving horizontally at 0.5 m/s.

As the drone moves, the photographer has to rotate the camera attached to the bottom of the drone to keep the bride and groom in view. When the drone is moving towards the bride and groom,

but is still 30 m away horizontally from them, at what rate would the camera need to be rotated to keep it aimed at the bride and groom? Measure the angle as the angle below horizontal, and include units in your answer.



Setup 1: θ below horizontal, $\tan\theta = \dfrac{y}{x}$

Solution

Variables.

- x: horizontal distance (m) between the drone and the bride and groom.
- y: height (m) of the drone above the ground.
- θ: angle (rad) of the camera below horizontal.

Given.

- Steady height: $y = 8$ m, so $\dfrac{dy}{dt} = 0$ m/s.
- The drone moves towards the bride and groom, so x is decreasing: $\dfrac{dx}{dt} = -0.5$ m/s.

Find. $\dfrac{d\theta}{dt}$ when $x = 30$ m.

Step 1: Relate θ to known quantities.

$$\tan\theta = \frac{y}{x}$$

Step 2: Differentiate with respect to t.

$$\frac{d}{dt}\big[\tan\theta\big] = \frac{d}{dt}\left[\frac{y}{x}\right]$$

Using the quotient rule,

$$\sec^2\theta \cdot \frac{d\theta}{dt} = \frac{\dfrac{dy}{dt}\,x - y\,\dfrac{dx}{dt}}{x^2}.$$

Since $\dfrac{dy}{dt} = 0$,

$$\sec^2\theta \cdot \frac{d\theta}{dt} = \frac{-y\,\dfrac{dx}{dt}}{x^2} \quad\Longrightarrow\quad \frac{d\theta}{dt} = \frac{-y\cos^2\theta \cdot \dfrac{dx}{dt}}{x^2}.$$

Step 3: Solve for θ.

$$\theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{8}{30}\right) \approx 0.2606 \text{ rad}$$

(Radians are required because the derivative formulas for trigonometric functions assume angles are measured in radians.)

Step 4: Substitute known values.

$$\frac{d\theta}{dt} = \frac{-(8)\cos^2(0.2606)\cdot(-0.5)}{30^2} = \frac{4\cos^2(0.2606)}{900} \approx 0.004 \text{ rad/s}$$

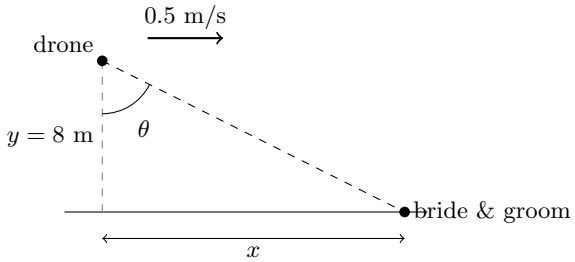
Answer. The camera would need to be rotated at approximately

$$\boxed{4 \times 10^{-3} \text{ rad/s}}$$

(the angle below horizontal is increasing as the drone approaches).

Alternative Solution

Here we instead measure the angle from the vertical, so the angle *decreases* as the drone approaches the bride and groom.



Setup 2: θ from vertical, $\tan\theta = \dfrac{x}{y}$

Variables.

- x: horizontal distance (m) between the drone and the bride and groom.
- y: height (m) of the drone above the ground.
- θ: angle (rad) between the vertical and the camera's line of sight.

Given.

- Steady height: $y = 8$ m, so $\dfrac{dy}{dt} = 0$ m/s.
- The drone moves towards the bride and groom, so x is decreasing: $\dfrac{dx}{dt} = -0.5$ m/s.

Find. $\dfrac{d\theta}{dt}$ when $x = 30$ m.

Step 1: Relate θ to known quantities.

$$\tan\theta = \frac{x}{y}$$

Step 2: Differentiate with respect to t.

$$\frac{d}{dt}\left[\tan\theta\right] = \frac{d}{dt}\left[\frac{x}{y}\right]$$

Using the quotient rule,

$$\sec^2\theta \cdot \frac{d\theta}{dt} = \frac{\dfrac{dx}{dt}\,y - x\,\dfrac{dy}{dt}}{y^2}.$$

Since $\dfrac{dy}{dt} = 0$,

$$\sec^2\theta \cdot \frac{d\theta}{dt} = \frac{\dfrac{dx}{dt}}{y} \quad \Longrightarrow \quad \frac{d\theta}{dt} = \frac{\cos^2\theta \cdot \dfrac{dx}{dt}}{y}.$$

Step 3: Solve for θ.

$$\theta = \tan^{-1}\left(\frac{x}{y}\right) = \tan^{-1}\left(\frac{30}{8}\right) \approx 1.31 \text{ rad}$$

(Again, radians are required.)

Step 4: Substitute known values.

$$\frac{d\theta}{dt} = \frac{\cos^2(1.31)\cdot(-0.5)}{8} \approx -0.004 \text{ rad/s}$$

Answer. The camera would need to be rotated at approximately

$$\boxed{-4 \times 10^{-3} \text{ rad/s}}$$

(the negative sign means the angle from the vertical is decreasing). This has the same magnitude as the first solution, since the two angles sum to $\pi/2$.

Problem 21

A cone made of sand is collapsing under its own weight, causing the height of the cone to decrease. None of the sand is blowing away, so **the volume of the cone remains constant**. As the height decreases, the base spreads out. The current height of the cone is 2 m and its base has a radius of 3 m. If the height of the cone is decreasing at a rate of 0.1 m/s, then how fast is the circumference of the base increasing?

The volume of a cone is given by $V = \frac{1}{3}\pi r^2 h$, where r is the radius of the circular base of the cone and h is its height.

Solution

Variables.

- V: volume (m^3) of the cone.
- r: radius (m) of the base.
- h: height (m) of the cone.
- C: circumference (m) of the base.

Given.

- The volume remains constant: $\dfrac{dV}{dt} = 0$ m^3/s.
- The base spreads out, so $\dfrac{dr}{dt} > 0$.
- Currently $h = 2$ m and $r = 3$ m.
- The height is decreasing: $\dfrac{dh}{dt} = -0.1$ m/s.

Find. $\dfrac{dC}{dt}$ at the current moment.

Step 1: Examine the formula for C.

$$C = 2\pi r$$

This involves only one variable, so to find $\dfrac{dC}{dt}$ we need $\dfrac{dr}{dt}$.

Step 2: Find a relationship that gives $\dfrac{dr}{dt}$. We know $\dfrac{dV}{dt}$ and $\dfrac{dh}{dt}$, so we use the volume formula

$$V = \frac{1}{3}\pi r^2 h$$

to solve for $\dfrac{dr}{dt}$, then work backwards to $\dfrac{dC}{dt}$.

Step 3: Solve for $\dfrac{dr}{dt}$. Differentiate both sides with respect to t, moving the constant out front:

$$\frac{d}{dt}[V] = \frac{1}{3}\pi \frac{d}{dt}\left[r^2 h\right].$$

Using the product rule,

$$\frac{dV}{dt} = \frac{1}{3}\pi \left[2r \frac{dr}{dt} h + r^2 \frac{dh}{dt}\right].$$

Substituting $\dfrac{dV}{dt} = 0$, $r = 3$, $h = 2$, and $\dfrac{dh}{dt} = -0.1$:

$$\begin{aligned} 0 &= \frac{1}{3}\pi\left[2(3)\frac{dr}{dt}(2) + 3^2(-0.1)\right] \\ 0 &= \frac{\pi}{3}\left[12\frac{dr}{dt} - 0.9\right] \\ 0.9 &= 12\frac{dr}{dt} \\ \frac{dr}{dt} &= \frac{0.9}{12} = 0.075 \text{ m/s}. \end{aligned}$$

Step 4: Use $\dfrac{dr}{dt}$ to solve for $\dfrac{dC}{dt}$.

$$\frac{d}{dt}[C] = \frac{d}{dt}[2\pi r] \quad \Longrightarrow \quad \frac{dC}{dt} = 2\pi\frac{dr}{dt} = 2\pi(0.075).$$

Answer. The circumference of the base is increasing at

$$\boxed{\frac{dC}{dt} = 0.15\pi \text{ m/s}} \approx 0.47 \text{ m/s}.$$

Alternative Solution 1: Substitute r from the circumference formula into the volume formula

Step 1: Rearrange the circumference formula for r.

$$C = 2\pi r \quad \Longrightarrow \quad r = \frac{C}{2\pi}.$$

Step 2: Substitute into the volume formula.

$$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi\left(\frac{C}{2\pi}\right)^2 h = \frac{C^2 h}{12\pi}.$$

Step 3: Differentiate with respect to t. Moving the constant out front and using the product rule,

$$\frac{dV}{dt} = \frac{1}{12\pi}\left[2C\frac{dC}{dt}h + C^2\frac{dh}{dt}\right].$$

Step 4: Substitute known values. Currently $r = 3$ m, so $C = 2\pi(3) = 6\pi$ m. Also $h = 2$ m, $\dfrac{dh}{dt} = -0.1$ m/s, and $\dfrac{dV}{dt} = 0$:

$$\begin{aligned} 0 &= \frac{1}{12\pi}\left[2(6\pi)(2)\frac{dC}{dt} + (6\pi)^2(-0.1)\right] \\ 0 &= \frac{1}{12\pi}\left[24\pi\frac{dC}{dt} - 3.6\pi^2\right] \\ 24\pi\frac{dC}{dt} &= 3.6\pi^2 \\ \frac{dC}{dt} &= \frac{3.6\pi}{24} = 0.15\pi \text{ m/s}. \end{aligned}$$

Answer.

$$\boxed{\frac{dC}{dt} = 0.15\pi \text{ m/s}} \approx 0.47 \text{ m/s}.$$

Alternative Solution 2: Substitute r from the volume formula into the circumference formula

Step 1: Find the constant volume. Since the volume never changes, we can compute it from the current values:

$$V = \frac{1}{3}\pi(3)^2(2) = 6\pi \text{ m}^3.$$

Step 2: Rearrange the volume formula for r.

$$V = \frac{1}{3}\pi r^2 h \quad \Longrightarrow \quad r^2 = \frac{3V}{\pi h} = \frac{3(6\pi)}{\pi h} = \frac{18}{h} \quad \Longrightarrow \quad r = \sqrt{\frac{18}{h}} = 3\sqrt{2}\, h^{-1/2},$$

taking the positive root since $r > 0$. (Check: at $h = 2$, $r = 3\sqrt{2}/\sqrt{2} = 3$.)

Step 3: Substitute into the circumference formula.

$$C = 2\pi r = 2\pi\left(3\sqrt{2}\, h^{-1/2}\right) = 6\sqrt{2}\,\pi\, h^{-1/2}.$$

Step 4: Differentiate with respect to t. Using the power rule and the chain rule,

$$\frac{dC}{dt} = 6\sqrt{2}\,\pi\left(-\frac{1}{2}\right) h^{-3/2}\,\frac{dh}{dt} = -3\sqrt{2}\,\pi\, h^{-3/2}\,\frac{dh}{dt}.$$

Step 5: Substitute known values. With $h = 2$ we have $h^{3/2} = 2\sqrt{2}$, and $\dfrac{dh}{dt} = -0.1$:

$$\frac{dC}{dt} = -\frac{3\sqrt{2}\,\pi}{2\sqrt{2}}\,(-0.1) = \frac{3\pi}{2}(0.1) = 0.15\pi \text{ m/s}.$$

Answer.

$$\boxed{\frac{dC}{dt} = 0.15\pi \text{ m/s}} \approx 0.47 \text{ m/s}.$$

Problem 22

A 15 cm × 8 cm rectangular piece of cardboard is to be made into an open-topped box by cutting from each corner a square of side length x, and then folding up the remaining edges; this will make a box of height x.

(a) Write out a formula for the volume of the box.
(b) What side length of square should be cut away to maximize the volume of the box? Compute the resulting volume as well, and include units in your answer.

Solution

Variable. Let x (cm) be the side length of the square cut from each corner. This is the decision variable: choosing it determines the shape of the box. The box has height x, width $8 - 2x$, and length $15 - 2x$.

(a) Volume formula.

$$\begin{aligned} V(x) &= (\text{length})(\text{width})(\text{height}) \\ &= (15 - 2x)(8 - 2x)(x) \\ &= (120 - 16x - 30x + 4x^2)\, x \\ &= \boxed{V(x) = 120x - 46x^2 + 4x^3} \end{aligned}$$

(b) Maximizing the volume.

The side lengths must be non-negative, so $8 - 2x \geq 0$, which gives the domain $0 \leq x \leq 4$.

To find the optimal x, differentiate and set the derivative equal to 0 to identify critical points:

$$\frac{dV}{dx} = 120 - 92x + 12x^2 = 0.$$

Dividing by 4,

$$3x^2 - 23x + 30 = 0.$$

By the quadratic formula,

$$x = \frac{23 \pm \sqrt{23^2 - 4(3)(30)}}{2(3)} = \frac{23 \pm 13}{6} \quad \Longrightarrow \quad x = 6 \;\text{ or }\; x = \frac{5}{3}.$$

Note (factoring instead). The quadratic can also be factored directly. We need two numbers that multiply to $3 \cdot 30 = 90$ and add to -23, which are -18 and -5. Splitting the middle term,

$$\begin{aligned} 3x^2 - 23x + 30 &= 3x^2 - 18x - 5x + 30 \\ &= 3x(x - 6) - 5(x - 6) \\ &= (3x - 5)(x - 6) = 0, \end{aligned}$$

so $x = \frac{5}{3}$ or $x = 6$, the same roots as the quadratic formula gives.

The root $x = 6$ is rejected: x can be at most $8/2 = 4$ for the cut squares to fit in the cardboard. Why? Think about the length $8 - 2x$. To be a real demension, it must be greater than or equal to

zero so the maximum length we can have can be solved by $8 - 2x = 0$ which yields a max of $x = 4$ So the only critical point in the domain is $x = \frac{5}{3} \approx 1.67$.

Check that this is a maximum. At the endpoints, $V(0) = 0$ and $V(4) = 0$, while $V\left(\frac{5}{3}\right) > 0$. (Also, $V''(x) = -92 + 24x$, so $V''\left(\frac{5}{3}\right) = -52 < 0$.)

Maximum volume.

$$V\left(\tfrac{5}{3}\right) = \left(15 - \tfrac{10}{3}\right)\left(8 - \tfrac{10}{3}\right)\left(\tfrac{5}{3}\right) = \frac{35}{3} \cdot \frac{14}{3} \cdot \frac{5}{3} = \frac{2450}{27} \approx 90.74\,\text{cm}^3.$$

Answer. Cutting squares of side length $x = \frac{5}{3} \approx 1.67\,\text{cm}$ from each corner produces the largest box, with a volume of approximately $\boxed{90.74\,\text{cm}^3}$.

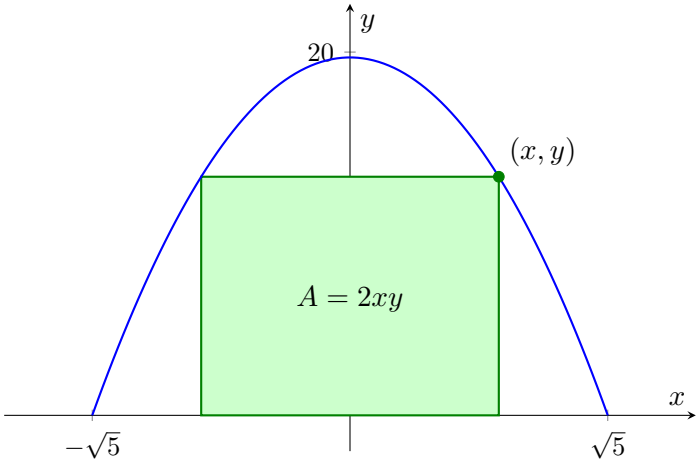
Problem 23

A rectangle is inscribed with its base on the x axis and its upper corners on the parabola $y = 20 - 4x^2$. What are the dimensions of such a rectangle with the greatest possible area? Compute the final rectangle area associated with those dimensions.

Solution

Sketch. The curve $y = 20 - 4x^2$ is a parabola opening downwards with vertex at $y = 20$. Its roots satisfy $4x^2 = 20$, so $x = \pm\sqrt{5}$.

Variables. Let (x, y) be the upper right corner of the rectangle, which lies on the parabola, with $x > 0$. By symmetry, the upper left corner is $(-x, y)$. Choosing x therefore defines any possible rectangle. The rectangle has width $2x$ and height $y = 20 - 4x^2$, and the domain is $0 < x < \sqrt{5}$.



Area function.

$$A(x) = 2x\,y = 2x\,(20 - 4x^2) = 40x - 8x^3.$$

Maximize. Take the derivative and set it equal to 0:

$$\frac{dA}{dx} = 2(20 - 4x^2) + 2x(-8x) = 40 - 24x^2 = 0.$$

Dividing by 2,

$$20 - 12x^2 = 0 \quad \Longrightarrow \quad x^2 = \frac{5}{3} \quad \Longrightarrow \quad x = \pm\sqrt{\frac{5}{3}}.$$

We take the positive root, since the area was defined with $x > 0$. Thus $x = \sqrt{5/3} \approx 1.29$, which lies in the domain $(0, \sqrt{5})$.

Check that this is a maximum. The area is 0 at both ends of the domain ($x = 0$ and $x = \sqrt{5}$) and positive in between, so the critical point gives the maximum. (Also, $A''(x) = -48x < 0$ for $x > 0$.)

Dimensions.

$$\text{width} = 2x = 2\sqrt{\tfrac{5}{3}} \approx 2.58,$$
$$\text{height} = y = 20 - 4\left(\tfrac{5}{3}\right) = \tfrac{40}{3} \approx 13.33.$$

Maximum area.

$$A\left(\sqrt{\tfrac{5}{3}}\right) = 2\sqrt{\tfrac{5}{3}}\left(20 - 4 \cdot \tfrac{5}{3}\right) = \frac{80}{3}\sqrt{\frac{5}{3}} \approx 34.43.$$

Answer. The rectangle of greatest area has width $2\sqrt{5/3} \approx 2.58$ and height $\frac{40}{3} \approx 13.33$, with area

$$\boxed{\frac{80}{3}\sqrt{\frac{5}{3}} \approx 34.43 \text{ square units}}.$$